



Job Aid: LRN DataLink CSV Data Preparation Guide OD2A Biosurveillance

Background: LRN DataLink allows you to submit multiple samples and test results in a single upload using a specifically formatted CSV file. This is NOT intended to replace Electronic Laboratory Reporting (ELR) via the LRN-defined HL7 2.5.1 standard.

Purpose: This document provides detailed instructions for creating the CSV file for upload through LRN DataLink for submitting OD2A biosurveillance results to CDC.

Audience: Laboratory personnel involved in submitting data to CDC.

Additional information for creating these files are as follows:

I. Data File Creation

- A. This CSV Data Preparation Guide is intended for use with **OD2A DataLink Template v02**.
- B. LRN DataLink is compatible with both Microsoft Edge and Google Chrome browsers.
- C. Download the "**Template with Example Data**" and use the specifications in **Table 1** of this document to create a compliant data file.
- D. Save the data file as a .csv with **no more than 1000 rows**, (i.e. header + 999 rows) for successful submission.
- E. Formatting:
 - a. The first row of the file must contain all headers in the order listed in the template and on **Table 1**.
 - b. **Delete** the Example_Data_1, Example_Data_2, Example_Data_3, Example_Data_4 and Example_Data_5 rows from the file.
 - c. If reusing a file **ensure all prior data is removed by deleting previously completed rows**, not just clearing the cell content. For example, if 5 rows of data were initially entered in the file but only 2 rows need to be submitted, the 3 rows that are not needed should be deleted.
 - d. **All 27 columns must be present**, even if you do not send data for the optional fields. Do not add any additional columns.
 - e. **Match** the template field names exactly. Write column headers exactly as they are in the guidelines and the template. For example, the Sample identifier header must be written as "Specimen_ID" to match the template.
 - f. DataLink will not accept files with missing or incorrect headers.
 - g. Any free text entry that consists of more than 11 characters and contains only numbers or begins with leading zeros should be enclosed in double quotes. This applies to Specimen_ID, Patient_ID, Sample_ID, Run_ID, Analyte_LOINC, Result_CODED,



Job Aid: LRN DataLink CSV Data Preparation Guide OD2A Biosurveillance

Result_NUMERIC, Unit_Of_Measure, Analyte_Result_Comment, Patient_County, Analyst, Reviewing_Official (Table 1).

- h. Patient identifiers should not be reused. When Patient_ID is unavailable, enter Specimen_ID in the Patient_ID column. For example, do not use placeholders like NA, N/A, ".", or repetitive entries such as sequential numbers (e.g., 1 to 20).
- i. Data in all rows with the same Specimen_ID must match except for the columns Analyte_LOINC, Test_dt, Result_CODED, and Analyte_Result_Comment.
- j. For an Analyte_LOINC, there can only be one Result_CODED value per Specimen_ID in the data file.
- k. A maximum of 150 Analytes can be added to a Specimen_ID when uploading an OD2A CSV file to DataLink.
- l. Duplicate rows in the data file shall be ignored during processing.

II. Data Updates, Corrections, and File Errors

- A. Results from testing multiple specimens can be included in the same file, whether they are Presumptive/Preliminary or Final results.
- B. Additional/Updated test results can be added for an already submitted specimen by uploading another file at a later date. **This file should include both the Presumptive/Preliminary results submitted and the Final results.** The data in the Preliminary and Final results rows should match, except for Analyte_LOINC, Test_dt, Result_CODED, and Analyte_Result_Comment.
- C. To correct a result for a previously submitted specimen, the original CSV file with the corrected data must be resubmitted, ensuring all previously sent data is completely replaced.
- D. File errors of 10 or less will display on the DataLink screen. If greater than 10 errors are present, an error report file will be available to download.

If you have any questions or concerns when sending data using this method, email LRNITServices@cdc.gov.



Job Aid: LRN DataLink CSV Data Preparation Guide OD2A Biosurveillance

Table 1: File Specifications

Column Header (field name)	CSV File Column	Description	Usage	Data Type	Validation Information
Activity_Type	A	Type of Activity	Required	Text	Must match one of the values in LRN Value Set table below. File will fail if acceptable value is not entered, or field is left blank.
Specimen_ID	B	Unique identifier for a single sample assigned by the Lab.	Required	Text	Maximum characters = 30 File will fail if this is left blank for any records.
Patient_ID	C	Unique patient identifier. This is typically the Medical Record Number. Do not send a Social Security Number.	Required	Text	Maximum characters = 20 File will fail if this is left blank for any records.
Sample_Type	D	Type of sample.	Required	Text	Must match one of the values in LRN Value Set table below. File will fail if acceptable value is not entered, or field is left blank.
Received_dt	E	Date and time sample received in laboratory.	Required	Time Stamp	Valid date format: MM/DD/YYYY HH:MM Enter time using 24-hour clock. File will fail if the value entered is not in valid date format or field is left blank.



Job Aid: LRN DataLink CSV Data Preparation Guide OD2A Biosurveillance

Column Header (field name)	CSV File Column	Description	Usage	Data Type	Validation Information
Collection_dt	F	Date and time sample was collected from patient.	Optional	Time Stamp	Valid date format: MM/DD/YYYY HH:MM Enter time using 24-hour clock. Invalid data will not be saved. An error message will not display.
Sample_ID	G	Unique identifier for a single sample assigned by the Submitter.	Optional	Text	Not Applicable for OD2A
Run_ID	H	A collection of samples	Optional	Text	Not Applicable for OD2A
Run_Create_dt	I	Date and time the run is created.	Optional	Text	Not Applicable for OD2A
Analyte_LOINC	J	LOINC code of the analyte.	Required	Text	Must match one of the values in LRN Value Set table below. File will fail if acceptable value is not entered, or field is left blank.
Test_dt	K	Date and time sample was tested.	Required	Time Stamp	Valid date format: MM/DD/YYYY HH:MM Enter time using 24-hour clock. File will fail if the value entered is not in valid date format or field is left blank.
Result_CODED	L	SNOMED code for the result for analyte.	Required	Text	Must match one of the values in LRN Value Set table below. File will fail if acceptable value is not entered, or field is left blank.
Result_Numeric	M	Numeric result for analyte.	Optional	Text	Not Applicable for OD2A
Unit_Of_Measure	N	Unit for Result_Numeri c	Optional	Text	Not Applicable for OD2A



Job Aid: LRN DataLink CSV Data Preparation Guide OD2A Biosurveillance

Column Header (field name)	CSV File Column	Description	Usage	Data Type	Validation Information
Analyte_Result_Comment	O	Enter comments from a physician or lab technician here. This field is not intended for characteristics of tests or statements about false positive or negative results.	Optional	Text	Maximum characters = 199 Text is truncated after the maximum allowable characters is exceeded.
Instrument_Type	P	Type of Instrument	Required	Text	Must match one of the values in LRN Value Set table below. File will fail if acceptable value is not entered, or field is left blank.
Patient_State	Q	Patient's state using 2-character abbreviation.	Optional	Text	Must match one of the values in LRN Value Set table below. -OR- Enter "NULL" -OR- Field can be left blank. Invalid data will not be saved. An error message will not display.
Patient_County	R	Patient's county.	Optional	Text	Maximum characters = 20 Text is truncated after the maximum allowable characters is exceeded.



Job Aid: LRN DataLink CSV Data Preparation Guide OD2A Biosurveillance

Column Header (field name)	CSV File Column	Description	Usage	Data Type	Validation Information
Patient_Zip	S	Patient's 5 or 9 digit zip code.	Optional	Text	Valid zip format: 99999 OR 99999-9999 -OR- Enter "NULL" -OR- Field can be left blank. Zip codes that begin with leading zeros should be enclosed in double quotes. Invalid data will not be saved. An error message will not display.
Patient_DOB	T	Patient's date of birth.	Optional	Time Stamp	Valid date format: MM/DD/YYYY -OR- Enter "NULL" -OR- Field can be left blank. Invalid data will not be saved. An error message will not display.
Patient_Age	U	Patient's age.	Optional	Text	Numeric value which is a whole number. Allows 4 characters. The maximum allowed Patient_Age is 199 years or 2,388 months, depending on the Age_Unit selected. -OR- Enter "NULL" -OR- Field can be left blank. Invalid data will not be saved. An error message will not display.



Job Aid: LRN DataLink CSV Data Preparation Guide OD2A Biosurveillance

Column Header (field name)	CSV File Column	Description	Usage	Data Type	Validation Information
Age_Unit	V	Patient age unit	Conditional	Text	Must match one of the values in LRN Value Set table below. -OR- Field can be left blank. File will fail if acceptable value is not entered, or field is left blank when Patient_Age is entered.
Patient_Sex	W	Patient's sex.	Optional	Text	Must match one of the values in LRN Value Set table below. -OR- Enter "NULL" -OR- Field can be left blank. Invalid data will not be saved. An error message will not display.
Patient_Race	X	Patient's race.	Optional	Text	Must match one of the values in LRN Value Set table below. -OR- Enter "NULL" -OR- Field can be left blank. If you send more than one, separate each value with "~". Invalid data will not be saved. An error message will not display.
Patient_Ethnicity	Y	Patient's ethnicity.	Optional	Text	Must match one of the values in LRN Value Set table below. -OR- Enter "NULL" -OR- Field can be left blank. If you send more than one, separate each value with "~". Invalid data will not be saved. An error message will not display.



Job Aid: LRN DataLink CSV Data Preparation Guide OD2A Biosurveillance

Column Header (field name)	CSV File Column	Description	Usage	Data Type	Validation Information
Analyst	Z	Name of Analyst.	Optional	Text	Not Applicable for OD2A
Reviewing_Official	AA	Name of Reviewing Official.	Optional	Text	Not Applicable for OD2A

III. Appendix: Value Sets

LRN Value Sets

Table 2: Activity Type

Activity Type
OD2A

Table 3: Sample Type

Sample Type
Blood
Plasma
Serum
Urine

Table 4: Analyte LOINC

Analyte LOINC		
Class	Name	LOINC Code
Benzodiazepines (Tier1)	Alprazolam [Presence] in Specimen	61038-6
	6-Alpha-Hydroxyalprazolam [Presence] in Specimen	LRNT144
	Clonazepam [Presence] in Specimen	61039-4
	7-Aminoclonazepam [Presence] in Specimen	61031-1
	Diazepam [Presence] in Specimen	61074-1
	Nordiazepam [Presence] in Specimen	61051-9
	Oxazepam [Presence] in Specimen	61055-0
	Temazepam [Presence] in Specimen	61061-8
	Lorazepam [Presence] in Specimen	61044-4
	Clonazolam [Presence] in Specimen	LRNT145
Benzodiazepines (Tier2)	Etizolam [Presence] in Specimen	LRNT146
	Flualprazolam [Presence] in Specimen	LRNT147
	Flubromazolam [Presence] in Specimen	LRNT148



Job Aid: LRN DataLink CSV Data Preparation Guide OD2A Biosurveillance

Analyte LOINC		
Class	Name	LOINC Code
Cannabinoids (Optional)	Delta-9 Tetrahydrocannabinol [Presence] in Specimen	LRNT149
	11-Hydroxy Delta-9 Tetrahydrocannabinol [Presence] in Specimen	48935-1
	Delta-9 Carboxy Tetrahydrocannabinol [Presence] in Specimen	61063-4
	Delta-8 Tetrahydrocannabinol [Presence] in Specimen	LRNT150
	11-Hydroxy Delta-8 Tetrahydrocannabinol [Presence] in Specimen	LRNT151
	Delta-8 Carboxy Tetrahydrocannabinol [Presence] in Specimen	LRNT152
	Delta-10 Tetrahydrocannabinol [Presence] in Specimen	LRNT153
	11-Hydroxy Delta-10 Tetrahydrocannabinol [Presence] in Specimen	LRNT154
	Delta-10 Carboxy Tetrahydrocannabinol [Presence] in Specimen	LRNT155
	Delta-9 Tetrahydrocannabinol-O-Acetate [Presence] in Specimen	LRNT156
	THCP (Tetrahydrocannabiphorol Isomers) [Presence] in Specimen	LRNT157
	2F-Deschloroketamine [Presence] in Specimen	LRNT158
Novel Stimulants/Hallucinogens (Tier2)	3-Hydroxyphencyclidine-Phencyclidine [Presence] in Specimen	LRNT159
	4-Hydroxyphencyclidine-Phencyclidine [Presence] in Specimen	LRNT160
	Alpha-Pyrrolidinohexiophenone [Presence] in Specimen	LRNT161
	Alpha-Pyrrolidinoisohexanophenone [Presence] in Specimen	LRNT162
	Eutylone [Presence] in Specimen	LRNT163
	N,N-Dimethyl Pentylone [Presence] in Specimen	LRNT164
	Alpha-Pyrrolidinohexiophenone/Alpha-Pyrrolidinoisohexanophenone [Presence] in Specimen	LRNT192
	3-Hydroxyphencyclidine-Phencyclidine/4-Hydroxyphencyclidine-Phencyclidine [Presence] in Specimen	LRNT193
Opioids (Tier1)	Fentanyl [Presence] in Specimen	61042-8
	Norfentanyl [Presence] in Specimen	61053-5
	Codeine [Presence] in Specimen	40626-4
	Codeine-6 β -D-Glucuronide [Presence] in Specimen	LRNT165
	Hydrocodone [Presence] in Specimen	32080-4
	Norhydrocodone [Presence] in Specimen	LRNT166
	Hydromorphone [Presence] in Specimen	32081-2
	Morphine [Presence] in Specimen	32100-0
	Morphine-3 β -D-Glucuronide [Presence] in Specimen	LRNT167
	Morphine-6 β -D-glucuronide [Presence] in Specimen	LRNT168
	Oxycodone [Presence] in Specimen	32101-8
	Noroxycodone [Presence] in Specimen	LRNT169
	Oxymorphone [Presence] in Specimen	LRNT170



Job Aid: LRN DataLink CSV Data Preparation Guide OD2A Biosurveillance

Analyte LOINC		
Class	Name	LOINC Code
	6-Monoacetylmorphine (6-MAM) [Presence] in Specimen	32099-4
	Methadone [Presence] in Specimen	32093-7
	2-Ethylidene-1,5-Dimethyl-3,3-Diphenylpyrrolidine (EDDP) [Presence] in Specimen	41859-0
	Buprenorphine [Presence] in Specimen	82373-2
	Norbuprenorphine [Presence] in Specimen	82375-7
	4-Anilino-N-Phenethylpiperidine [Presence] in Specimen	LRNT171
	Acetylfentanyl [Presence] in Specimen	LRNT172
	Fluorofentanyl [Presence] in Specimen	LRNT173
	Para-Fluorofentanyl (P-FF) [Presence] in Specimen	LRNT174
	Ortho-Fluorofentanyl (O-FF) [Presence] in Specimen	LRNT175
	Meta-Fluorofentanyl (M-FF) [Presence] in Specimen	LRNT176
	Carfentanil [Presence] in Specimen	LRNT177
	Norcarfentanil [Presence] in Specimen	LRNT178
	Tramadol [Presence] in Specimen	LRNT179
	O-Desmethyltramadol [Presence] in Specimen	LRNT180
Opioids (Tier 2)	2-Methyl-AP-237 [Presence] in Specimen	LRNT181
	Brorphine [Presence] in Specimen	LRNT182
	Etodesnitazene [Presence] in Specimen	LRNT183
	Isotonitazene [Presence] in Specimen	LRNT184
	Metonitazene [Presence] in Specimen	LRNT185
	N-Pyrrolidino Etonitazene [Presence] in Specimen	LRNT186
	Protonitazene [Presence] in Specimen	LRNT187
	Isotonitazene/Protonitazene [Presence] in Specimen	LRNT194
Other Substances (Tier 1)	Xylazine [Presence] in Specimen	LRNT188
	Gabapentin [Presence] in Specimen	LRNT189
Stimulants (Tier 1)	Amphetamines [Presence] in Specimen	29530-3
	Methamphetamine [Presence] in Specimen	40381-6
	Cocaine [Presence] in Specimen	40625-6
	Benzoyllecgonine [Presence] in Specimen	40609-0
	Cocaethylene [Presence] in Specimen	LRNT195
Stimulants (Tier 2)	Methylenedioxyamphetamine [Presence] in Specimen	61046-9
	Methylenedioxymethamphetamine [Presence] in Specimen	40481-4
	Methylenedioxyethylamphetamine [Presence] in Specimen	61049-3

Note: Enter LOINC code for Analyte LOINC column in the CSV file.

Table 5: Result CODED

Result CODED	
Result	Result SNOMED Code
Detected	260373001
Not detected	260415000



Job Aid: LRN DataLink CSV Data Preparation Guide OD2A Biosurveillance

Result CODED	
Equivocal result	280414007
Inconclusive	419984006
Test not done	373121007

Note: Enter Result SNOMED Code for Result CODED column in the CSV file.

Table 6: Instrument Type

Instrument Type
LC/MS/MS
ICP/MS
GC/MSD
LC/ICP/MS
GC/MS/MS
LC/HRMS

Table 7: Patient State

Patient State			
AL	IL	MT	PR
AK	IN	NE	RI
AZ	IA	NV	SC
AR	KS	NH	SD
CA	KY	NJ	TN
CO	LA	NM	TX
CT	ME	NY	UT
DC	MD	NC	VT
DE	MA	ND	VA
FL	MI	OH	WA
GA	MN	OK	WV
HI	MS	OR	WI
ID	MO	PA	WY

Table 8: Age Unit, Patient Sex, Race, and Ethnicity

Age Unit	Patient Sex	Patient Race	Patient Ethnicity
Hour	M	American Indian or Alaska Native	Hispanic
Day	F	Asian	Not-Hispanic



Job Aid: LRN DataLink CSV Data Preparation Guide OD2A Biosurveillance

Age Unit	Patient Sex	Patient Race	Patient Ethnicity
Month		Black or African American	Unknown
Year		Native Hawaiian or Other Pacific Islander	
		White	
		Other Race	

IV. Revision History

Revision	Brief Description of Change	Effective Date
01	- The section on Data Updates, Corrections, and File Errors has been revised to include information on sending Presumptive/Preliminary and Final results. - A new column titled "CSV File Column" has been added to Table 1.	10/31/2024
02	- The following analytes have been added to Table 4: Analyte LOINC - o Alpha-Pyrrolidinohexiophenone/Alpha-Pyrrolidinoisohexanophenone [Presence] in Specimen - o 3-Hydroxyphencyclidine-Phencyclidine/4-Hydroxyphencyclidine-Phencyclidine [Presence] in Specimen - o Isotonitazene/Protonitazene [Presence] in Specimen - The analyte "Cocaethylene [Mass/volume] [Presence] in Specimen" (LOINC code 48946-8) under the Stimulants (Tier 1) class has been updated to "Cocaethylene [Presence] in Specimen" (LOINC code LRNT195).	11/18/2024
03	- Added details on deleting existing data in the file under Section 1: Data File Creation. - Maximum number of Analytes that can be added to the Specimen_ID has been increased from 100 to 150 in Section 1: Data File Creation.	12/20/2024
04	- Updated the Date File Creation section to specify the use of the Data Preparation Guide with the OD2A DataLink Template V02. - Updated the total number of columns in the OD2A template to 27. - Removed details from Table 1 and Table 8 in the Data Preparation Guide for 2 data elements. - CSV File Column in Table 1 has been updated to match the OD2A DataLink template v02. - Added the maximum allowed entry for Patient_Age when Age_Unit is Month in Table 1.	01/31/2025
05	- Updated Data File Creation section to include guidelines for patient identifier (Patient_ID) entry.	02/17/2025



Job Aid: LRN DataLink CSV Data Preparation Guide OD2A Biosurveillance

	- Updated Patient_Sex description in Table 1. - Updated Value Set list for Data Element Patient_Sex in Table 8.	
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